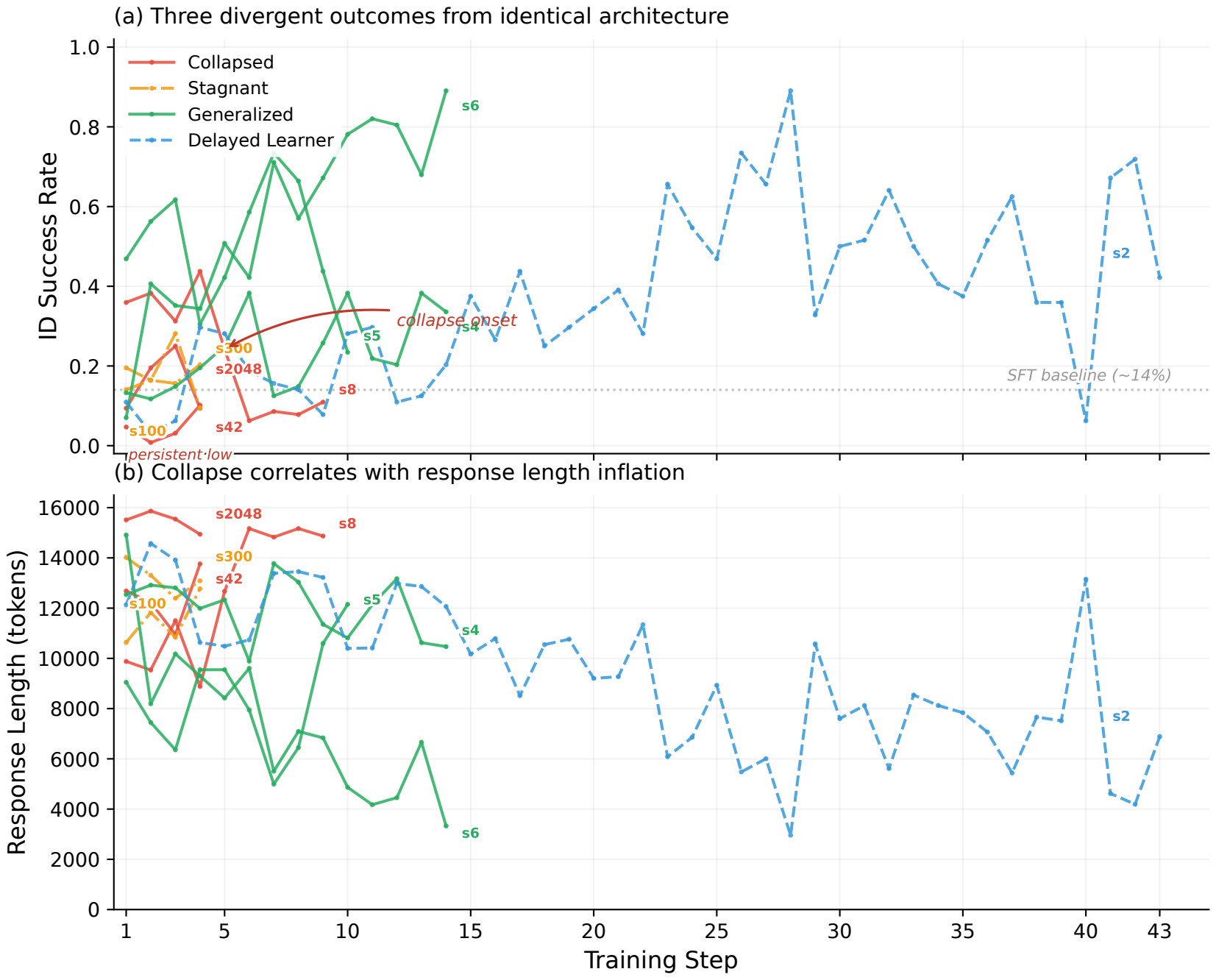

(a) Three divergent outcomes from identical architecture
Collapsed
Stagnant
Generalized
Delayed Learner
ID Success Rate
1.0
0.8
0.6
0.4
0.2
0.0
s6
s2
s4
s5
s300
s2048
s8
s100
s42
collapse onset
SFT baseline (~14%)
persistent·low
(b) Collapse correlates with response length inflation
Response Length (tokens)
16000
14000
12000
10000
8000
6000
4000
2000
0
s2048
s8
s300
s42
s100
s5
s4
s6
s2
1
5
10
15
20
25
30
35
40
43
Training Step